Free-body diagrams

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Learning target: Draw only the external forces on one chosen object.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 12.*
Mass 2 kg; new acceleration 5 m/s². At fixed mass, acceleration scales with net force.

02. Worked example: explain the model
Tension up and weight down; lamp-on-cable force acts on a different object.

03. A book at rest
Equal normal up and weight down; table and Earth.

04. After the hand lets go
Weight down, normal up, kinetic friction left; no hand force after release.

05. A falling ball
Weight down longer than upward drag; net force down.

06. Review: connect an earlier idea *Review: Unit 1, Day 3.*
v=(6-18)/4=-3 m/s, left. Straight descending graph between the stated endpoints.

07. Exit check
Only Earth’s weight down. Hand no longer contacts; upward velocity not a force.

08. STRETCH: test the model
Both diagrams can have equal tension and weight. Force balance alone does not tell velocity.

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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 12.*
Mass=1/slope≈0.6667 kg; a=3.75 m/s². Resistance subtracts from driving force, producing an offset.

02. Worked example: explain the model
Tension up and weight down; lamp-on-cable force acts on a different object.

03. Motionless lamp
Tension 19.6 N up; weight 19.6 N down.

04. A rough ramp
Weight vertically down; normal perpendicular out; static friction up slope opposes gravity’s downslope tendency.

05. Critique the extra arrows
Remove velocity force. Show net force separately as a sum, not an additional interaction. Check friction direction and actual sources.

06. Review: connect an earlier idea *Review: Unit 1, Day 3.*
v=6 m/s; position increases 6 m each second. x(8)=32 m.

07. Exit check
Weight down, normal up, friction right. No vertical acceleration (equal vertical forces); horizontal acceleration right if friction only horizontal force.

08. STRETCH: test the model
2 T cos 30°=40 N, so T≈23.1 N. Horizontal components cancel; do not set each tension to 20 N.